ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B. Tech. (CSE) (CC)

Semester: IV

SUBJECT CODE: BCSI404

COMPUTER NETWORKS

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. (10 Marks)

Q. 1. Define the following terms: (1 × 5 = 5 Marks)

(i) Network Topology

(ii) IP Address

(iii) OSI Model

(iv) Hub

(v) WAN

Q. 2. Answer the following questions: (1 × 5 =5 Marks)

(i) What are the functions of a Router?

(ii) Explain the difference between TCP and UDP.

(iii) What is the use of the ARP protocol?

(iv) What are the advantages of wireless networks?

(v) Define CIDR and its purpose.

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i) (a) Describe different types of network topologies with diagrams.

OR

(b) Compare OSI and TCP/IP models with examples.

(ii) (a) Explain the function of Ethernet and its standards.

OR

(b) Differentiate between Switch and Router with suitable examples.

Q.4.(i)(a) Write a note on IPv4 and IPv6 addressing formats.

OR

(b) Explain the use and benefits of the Internet Control Message Protocol (ICMP).

(ii)(a) What are the causes of network failures and how can they be identified?

OR

(b) Describe the steps involved in network troubleshooting using CLI tools.

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20Marks)

Q.5.(a) Explain in detail the various WAN technologies and their applications.

OR

(b) Describe tunneling and encryption protocols used in network Security.

Q.6.(a) Discuss the troubleshooting model for networks and explain any four troubleshooting tools with examples.

OR

(b) Design a small LAN for your department with suitable topology, devices, addressing, and media. Justify your choices.